

Name: _____

Period: _____

Date: _____

Topic 2.4 Guided Notes: Encoding Memories

Unit 2: Cognition | AP Psychology

Directions: Read through the topic overview page below and fill in each blank as you go. Use the tables and activities to check your understanding, then head to the quiz link at the end to test yourself.

Source page: appsychreviewhub.com/overviews/topic2.4

Word Bank

<i>Chunking</i>	<i>Categories</i>	<i>Primacy Effect</i>
<i>Spacing Effect</i>	<i>Massed Practice</i>	<i>Memory Consolidation</i>
<i>Distributed Practice</i>	<i>Mnemonic Devices</i>	<i>Serial Position Effect</i>
<i>Method of Loci</i>	<i>Hierarchies</i>	<i>Recency Effect</i>

1. Organizing the Chaos

Your short-term memory is incredibly limited, holding only about seven items at a time. To bypass this limit, your brain needs to organize raw data into meaningful structures.

- _____: a strategy that involves grouping individual, isolated pieces of information into larger, more meaningful units. Think about a phone number: you don't memorize ten random digits — you group it into three distinct parts, and suddenly ten items become three easy-to-remember chunks.
- _____ are mental groupings of similar objects or ideas. _____ take this a step further by arranging information in levels, from broad concepts down to specific subcategories (think of how a biology textbook organizes life from Domain down to Species).

Apply It: Which Organizing Strategy?

Using the word bank above, identify which concept best matches each scenario.

#	Scenario	Concept
1	A biology textbook organizes all life from the broad Domain level down to the specific Species level.	
2	A student groups all her flashcards about muscles into one pile and all her flashcards about bones into a separate pile.	

2. Mental Gymnastics: Mnemonic Devices

_____ are memory aids that help you organize information in a quirky, meaningful, or highly visual way, like using acronyms or rhymes.

One of the oldest and most powerful mnemonic devices is the _____ (also known as a Memory Palace). This technique involves associating the information you need to memorize with specific physical locations in a familiar place — you might mentally picture a grocery list scattered around rooms in your own house, then mentally "walk" through the house to retrieve the items during a test.

Apply It: Which Memory Aid?

#	Scenario	Concept
1	To remember the order of operations in math, a student uses the phrase "Please Excuse My Dear Aunt Sally."	
2	To memorize his speech, a man mentally places each talking point in a different room of his childhood home, then imagines walking through the house during his presentation.	

3. The Power of Timing: Cramming vs. Consistency

How you space out your studying actually changes how your brain physically stores the memory.

- _____: studying all the information in one single, grueling session (also known as cramming). While it might help you pass a quiz the next morning, the information vanishes from your brain almost immediately afterward.
- _____: spreading your study sessions out over a longer period of time (e.g., studying 20 minutes a day for a week instead of 3 hours the night before).

Psychologists have proven that distributed practice leads to significantly better long-term retention — a phenomenon known as the _____. Why does this work? Because it gives your brain time for _____ — the biological process of stabilizing and strengthening a memory trace, transferring it securely into long-term storage. A massive part of this happens while you sleep. If you pull an all-nighter to cram, you are actively depriving your brain of the sleep it needs!

Apply It: Which Timing Concept?

#	Scenario	Concept
1	A student pulls an all-nighter, reading his entire textbook cover to cover in one sitting the night before the final exam.	
2	While sleeping, a student's brain stabilizes and strengthens the memory trace of material she studied earlier that day, transferring it into long-term storage.	

4. The Order Matters: The Serial Position Effect

Encoding is also heavily affected by the order in which information is presented to you. Imagine your friend verbally gives you a 15-item grocery list, but you don't have a pen to write it down. By the time you get to the store, which items will you remember?

The _____ predicts that you will easily remember the items at the beginning and the end of the list, but you will completely forget the stuff in the middle. This happens for two distinct reasons:

- _____: you remember the items at the beginning of the list because you had time to mentally rehearse them over and over, successfully moving them into your long-term memory.
- _____: you remember the items at the end of the list because they were the last things you heard, meaning they are still freshly bouncing around in your short-term working memory.

Apply It: Primacy or Recency?

#	Scenario	Effect
1	A student can easily recall the first few vocabulary words her teacher read aloud in class because she had extra time to silently repeat them to herself.	
2	A student can easily recall the last few vocabulary words her teacher read aloud, since she just heard them moments ago.	

5. Don't Trip Up! (Common Misconceptions)

_____ is the action or strategy you choose to do (studying a little bit every day). The _____ is the psychological outcome (the fact that you actually remembered the information better on test day because of your strategy).

_____ works because of rehearsal (long-term memory). _____ works because the info is still fresh in your working memory.

6. Quick Check: Review Questions

Circle the best answer for each question below.

1. Instead of trying to memorize a 10-digit phone number as ten separate, isolated digits, a woman groups them into three smaller units, like 555-867-5309, making the number far easier to remember. This strategy is called:

- A. The Serial Position Effect
- B. The Method of Loci
- C. Massed Practice
- D. Chunking

2. To memorize the colors of the rainbow in order, a student uses the acronym ROY G. BIV, where each letter represents the first letter of a color. This memory aid, which organizes information in a quirky and memorable way, is called a:
- A. The Method of Loci
 - B. Hierarchy
 - C. Distributed Practice
 - D. Mnemonic Device
3. Instead of cramming the night before her exam, a student studies her vocabulary flashcards for just 20 minutes every single day over the course of two weeks leading up to the test. This strategy of spreading out study sessions over time is called:
- A. The Method of Loci
 - B. Chunking
 - C. Distributed Practice
 - D. Massed Practice
4. A researcher finds that students who studied vocabulary in short sessions spread out over two weeks scored significantly higher on a final exam than students who crammed the same total number of study hours into one night. This resulting improvement in long-term retention, caused by spacing out study sessions, is best described as:
- A. Distributed Practice
 - B. The Serial Position Effect
 - C. Massed Practice
 - D. The Spacing Effect
5. A friend recites a 15-item grocery list out loud, and a woman later realizes she has almost completely forgotten items 6 through 10, the ones recited in the middle of the list. According to the Serial Position Effect, why did she likely forget these specific items?
- A. Middle items are always semantically encoded, making them less memorable than the other items
 - B. Massed practice specifically targets and erases only the middle portion of any memorized list
 - C. They had already faded from short-term memory but weren't rehearsed long enough to reach long-term memory
 - D. The middle items were spoken in a much quieter, harder-to-hear tone of voice



Check Your Progress

Done with your notes? Head over to the [Quiz](https://appspsychreviewhub.com/quiz) to test what you've learned:

<https://appspsychreviewhub.com/quiz>